The Origin and Propagation of Noise in Gene Expression

Quantitative Biology Lecture Notes

This lecture note is based primarily on Johan Paulsson's 2004 Nature paper, "*Summing up the noise in gene networks*," together with its Supplementary Information (SI). The paper investigates the origins of stochastic fluctuations in gene expression, how these fluctuations propagate through the central dogma, and how they are affected by autorepression.

Here, we derive Paulsson's noise formula (Eq. (1)) starting from a birth–death Markov process. Our derivation also draws on the Linear Noise Approximation approach developed by Johan Elf and Måns Ehrenberg in "*Fast Evaluation of Fluctuations in Biochemical Networks With the Linear Noise Approximation*." Equations labeled $S1, S2, \ldots$ correspond to the notation used in Paulsson's Supplementary Information.

1 Noise in Gene Expression

Random fluctuations in genetic networks are inevitable as chemical reactions are probabilistic, and many genes, RNAs, and proteins occur in small numbers per cell. Roughly speaking, we can call random deviations from what we might expect in a deterministic view of gene expression **stochasticity**, or **noise**.

1.1 Intrinsic and Extrinsic Noise

All cell components display **intrinsic noise** due to random births and deaths of individual molecules, and **extrinsic noise** due to fluctuations in reaction rates. The terms 'intrinsic' and 'extrinsic' generically distinguish between the origin and propagation of noise, and their biological meaning is always defined in conjunction with a specified component or process.

For example, if a gene for a transcriptional repressor spontaneously switches on and off, thereby enslaving the encoded protein and transmitting the fluctuations to repressed genes, the noise is intrinsic to the number of active repressor genes and extrinsic to all affected components.

1.2 Measures of Gene-Expression Noise

For a molecular count n_i with mean $\langle n_i \rangle$, we use σ_i to denote the standard deviation, and σ to denote the covariance matrix:

$$\sigma_i^2 = \langle (n_i - \langle n_i \rangle)^2 \rangle, \qquad \sigma_{ij} = \langle (n_i - \langle n_i \rangle)(n_j - \langle n_j \rangle) \rangle, \qquad \sigma_{ii} = \sigma_i^2.$$

Paulsson compares the relative variance and the Fano factor:

$$\text{Relative variance} = \frac{\sigma_i^2}{\langle n_i \rangle^2}$$

$$\text{Fano factor} = \frac{\sigma_i^2}{\langle n_i \rangle}$$

The two measures therefore differ by a factor of $\langle n_i \rangle$. For a Poisson distribution, the Fano factor equals one. Paulsson expresses equation (1) in terms of relative variance, whose use we explain after the derivation.

2 A Single-Species Birth–Death Process

Chemical reactions in single cells in general occur far from thermodynamic equilibrium, and the molecule copy numbers can sometimes be very small. Both these properties make it mandatory to analyze the stochastic behavior of chemical reactions in living systems. One technique is to solve the **chemical Master equation**, which describes the time evolution of the probability distribution for all the different molecular components in the system.

We first consider the simplest single-species birth–death process,

$$\varnothing \xrightarrow{birth} X \xrightarrow{death} \varnothing$$

$$\varnothing \xrightarrow{k} X, \qquad X \xrightarrow{\gamma} \varnothing$$

Here k is the production rate, while each molecule is degraded independently at rate γ. If there are n molecules of X, the total degradation rate is therefore γn.

At the macroscopic level, the molecule number satisfies

$$\frac{dn}{dt} = k - \gamma n$$

The deterministic stationary copy number is

$$n^* = \frac{k}{\gamma}$$

To describe the stochastic dynamics, let $P_n(t)$ denote the probability of having n molecules at time t. The possible transitions are

$$n-1 \xleftarrow{\gamma n} n \xrightarrow{k} n+1$$

The corresponding Master equation is

$$\frac{dP_n}{dt} = kP_{n-1} + \gamma(n+1)P_{n+1} - kP_n - \gamma n P_n$$

The first two terms are probability inflows to state n, while the last two terms are probability outflows from state n.

2.1 Master Equation and Stationary Solution

We now look for the stationary probability distribution, for which

$$\frac{dP_n}{dt} = 0$$

The birth–death process can be represented as transitions between neighboring molecule-number states,

$$P_0 \underset{\gamma}{\overset{k}{\rightleftharpoons}} P_1 \underset{2\gamma}{\overset{k}{\rightleftharpoons}} P_2 \underset{3\gamma}{\overset{k}{\rightleftharpoons}} \cdots$$

The forward transition from state n to state $n+1$ occurs with rate k, whereas the reverse transition occurs with rate $\gamma(n+1)$.

We define the net **probability flux** from state n to state $n+1$ as

$$J_n = kP_n - \gamma(n+1)P_{n+1}$$

The Master equation can then be written in terms of the probability fluxes as

$$\frac{dP_n}{dt} = kP_{n-1} + \gamma(n+1)P_{n+1} - kP_n - \gamma nP_n = J_{n-1} - J_n$$

At stationarity, we have

$$J_{n-1} = J_n$$

Thus the probability flux is the same between every pair of neighboring states,

$$J_0 = J_1 = J_2 = \cdots$$

For a normalizable stationary distribution, there can be no finite probability current flowing to arbitrarily large molecule numbers. Hence, in the limit $n \to \infty$,

$$J_n \to 0$$

Since the stationary flux is independent of n, this implies

$$J_n = 0 \qquad \text{for all } n$$

Therefore,

$$kP_n = \gamma(n+1)P_{n+1} \tag{2.1}$$

This is the **detailed balance condition**: at stationarity, the probability flux from n to $n+1$ is exactly balanced by the reverse flux from $n+1$ to n.

Equation (2.1) gives

$$P_{n+1} = \frac{k}{\gamma(n+1)}P_n$$

Applying this relation successively from state 0 to state n gives

$$\begin{aligned} P_n &= P_0 \prod_{m=1}^{n} \frac{k}{\gamma m} \\ &= P_0 \frac{1}{n!}\left(\frac{k}{\gamma}\right)^n \end{aligned} \tag{2.2}$$

The probability distribution must satisfy

$$\sum_{n=0}^{\infty} P_n = 1$$

Using Eq. (2.2),

$$\begin{aligned} 1 &= P_0 \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{k}{\gamma}\right)^n \\ &= P_0 \exp\left(\frac{k}{\gamma}\right) \end{aligned}$$

Therefore,

$$P_0 = \exp\left(-\frac{k}{\gamma}\right)$$

and the stationary probability distribution follows a **Poisson distribution**,

$$P_n = \exp\left(-\frac{k}{\gamma}\right) \frac{(k/\gamma)^n}{n!} \sim \text{Poisson}\left(\frac{k}{\gamma}\right)$$

Its mean and variance are

$$\langle n \rangle = \frac{k}{\gamma}, \qquad \sigma^2 = \frac{k}{\gamma}$$

The **coefficient of variation** and **Fano factor** are therefore

$$\text{CV} = \frac{\sigma}{\langle n \rangle} = \frac{1}{\sqrt{\langle n \rangle}},$$

$$F = \frac{\sigma^2}{\langle n \rangle} = 1$$

Thus, although the absolute variance increases with expression level, the relative fluctuations decrease as the average molecule number increases.

3 Linear Noise Approximation

The Master equation can be written directly from the rate constants and stoichiometries of all the elementary reactions, but analytical and numerical solutions are generally difficult to obtain. It can often be simplified through van Kampen's Ω-expansion, which gives a **Linear Noise Approximation** (LNA) near macroscopic trajectories or stationary points.

3.1 Macroscopic and Mesoscopic Dynamics

We now consider a single chemical species X in a system of volume Ω. The molecule number is denoted by n, and the macroscopic concentration by x.

Consider a general single-species birth–death process,

$$\varnothing \xrightarrow{R_+(n)} X, \qquad X \xrightarrow{R_-(n)} \varnothing.$$

The possible transitions are

$$n-1 \xleftarrow{R_-(n)} n \xrightarrow{R_+(n)} n+1$$

A birth event changes the molecule number by $+1$, while a death event changes it by -1.

In the macroscopic limit,

$$x = \lim_{\Omega\to\infty} \frac{n}{\Omega}$$

and the transition rates per unit volume approach the macroscopic reaction rates

$$r_{\pm}(x) = \lim_{\substack{\Omega\to\infty \\ n/\Omega\to x}} \frac{R_{\pm}(n;\Omega)}{\Omega}$$

The corresponding Master equation for the molecule-number probability $P_n(t)$ is

$$\frac{\partial P_n}{\partial t} = R_+(n-1)P_{n-1} + R_-(n+1)P_{n+1} - [R_+(n) + R_-(n)]\,P_n \tag{3.1}$$

Define the shift operators

$$E^{\pm} f(n) = f(n \pm 1)$$

Equation (3.1) can then be written as

$$\frac{\partial P}{\partial t} = (E^- - 1)(R_+ P) + (E^+ - 1)(R_- P) \tag{3.2}$$

We now show how the macroscopic dynamics and the fluctuations around them emerge from the Master equation when the system volume goes to infinity.

3.2 System-Size Expansion to the Linear Fokker–Planck Equation

The Ω-expansion Taylor-expands the Master equation near macroscopic trajectories or stationary solutions in powers of $\Omega^{-1/2}$. The leading-order terms reproduce the macroscopic rate equation, and the next order terms give the Linear Noise Approximation.

To separate the macroscopic dynamics from fluctuations, define

$$n(t) = \Omega x(t) + \sqrt{\Omega}\,\xi(t)$$

Equivalently,

$$\frac{n}{\Omega} = x + \Omega^{-1/2}\xi$$

Let $\Pi(\xi, t)$ denote the probability density of the fluctuation variable ξ. The probability distribution $P_n(t)$ is related to $\Pi(\xi, t)$ through the change of variables,

$$\frac{\partial P_n}{\partial t} = \frac{\partial \Pi}{\partial t} + \frac{\partial \Pi}{\partial \xi}\frac{\partial \xi}{\partial t} = \frac{\partial \Pi}{\partial t} - \sqrt{\Omega}\,\dot{x}\frac{\partial \Pi}{\partial \xi} \tag{3.3}$$

Since at fixed molecule number

$$\frac{\partial n}{\partial t} = \Omega\dot{x} + \sqrt{\Omega}\frac{\partial \xi}{\partial t} = 0$$

gives

$$\frac{\partial \xi}{\partial t} = -\sqrt{\Omega}\,\dot{x}$$

Taylor expansion of the mesoscopic transition rates about the macroscopic concentration x:

$$\begin{aligned}\Omega^{-1}R_{\pm}(\Omega x+\sqrt{\Omega}\xi;\Omega) &= r_{\pm}\left(x+\Omega^{-1/2}\xi\right)\\ &= r_{\pm}(x)+\Omega^{-1/2}\xi r'_{\pm}(x)+O(\Omega^{-1})\end{aligned} \tag{3.4}$$

A change of one molecule corresponds to

$$n\to n\pm 1 \qquad\Longleftrightarrow\qquad \xi\to\xi\pm\Omega^{-1/2}$$

The shift operators can therefore be Taylor-expanded as

$$\begin{aligned}E^{\pm}f(\xi) &= f(\xi\pm\Omega^{-1/2})\\ &= 1\pm\Omega^{-1/2}\frac{\partial}{\partial\xi}+\frac{1}{2}\Omega^{-1}\frac{\partial^2}{\partial\xi^2}+O(\Omega^{-3/2})\end{aligned} \tag{3.5}$$

Substituting Eqs. (3.3), (3.4), and (3.5) into the Master equation (3.2) gives

$$\begin{aligned}\frac{\partial p_n}{\partial t} &= \frac{\partial\Pi}{\partial t}-\sqrt{\Omega}\,\dot{x}\frac{\partial\Pi}{\partial\xi}\\ &= \left[-\Omega^{-1/2}\frac{\partial}{\partial\xi}+\frac{1}{2}\Omega^{-1}\frac{\partial^2}{\partial\xi^2}+O(\Omega^{-3/2})\right]\times\Omega\left[r_+(x)+\Omega^{-1/2}\xi r'_+(x)+O(\Omega^{-1})\right]\Pi\\ &\quad+\left[\Omega^{-1/2}\frac{\partial}{\partial\xi}+\frac{1}{2}\Omega^{-1}\frac{\partial^2}{\partial\xi^2}+O(\Omega^{-3/2})\right]\times\Omega\left[r_-(x)+\Omega^{-1/2}\xi r'_-(x)+O(\Omega^{-1})\right]\Pi\end{aligned}$$

The differential operators act on all factors to their right.

Identifying terms of order $\Omega^{1/2}$ gives

$$-\sqrt{\Omega}\,\dot{x}\frac{\partial\Pi}{\partial\xi}=-\sqrt{\Omega}\left[r_+(x)-r_-(x)\right]\frac{\partial\Pi}{\partial\xi}$$

These terms cancel when

$$\dot{x}=r_+(x)-r_-(x)$$

which is exactly the macroscopic rate equation.

Identifying terms of order Ω^0 gives

$$\frac{\partial\Pi}{\partial t}=-\left[r'_+(x)-r'_-(x)\right]\frac{\partial}{\partial\xi}(\xi\Pi)+\frac{1}{2}\left[r_+(x)+r_-(x)\right]\frac{\partial^2\Pi}{\partial\xi^2} \tag{3.6}$$

Define

$$A=\frac{d}{dx}\left[r_+(x)-r_-(x)\right]=r'_+(x)-r'_-(x),$$
$$B^2=r_+(x)+r_-(x)$$

Equation (3.6) becomes

$$\frac{\partial\Pi}{\partial t}=-\frac{\partial}{\partial\xi}(A\xi\Pi)+\frac{1}{2}B^2\frac{\partial^2\Pi}{\partial\xi^2} \tag{3.7}$$

This is the **linear Fokker–Planck equation**. Its drift is linear in the fluctuation ξ, while A and B^2 depend on time through the macroscopic trajectory $x(t)$. Terms of order $\Omega^{-1/2}$ and smaller have been discarded.

3.3 The Stochastic Differential Equation

The linear Fokker–Planck equation (3.7) can equivalently be formulated as the linear Langevin equation (or the stochastic differential equation) as

$$d\xi = A\xi\, dt + B\, dW(t),$$

where $W(t)$ is a standard Wiener process satisfying

$$\langle dW \rangle = 0, \qquad \langle dW^2 \rangle = dt$$

3.4 Stationary Gaussian Fluctuations

Define the probability current as

$$J(\xi) = A\xi\Pi - \frac{B^2}{2}\frac{\partial \Pi}{\partial \xi}$$

The linear Fokker–Planck equation (3.7) can then be written as

$$\frac{\partial \Pi}{\partial t} = -\frac{\partial J}{\partial \xi}$$

At stationarity, $\partial\Pi/\partial t = 0$, so J is constant. For a normalizable distribution with vanishing probability flux at infinity, $J = 0$, giving

$$A\xi\Pi = \frac{B^2}{2}\frac{\partial \Pi}{\partial \xi}$$

Separating variables gives

$$\frac{d\Pi}{\Pi} = \frac{2A}{B^2}\xi\, d\xi$$

Integrating both sides,

$$\int \frac{d\Pi}{\Pi} = \frac{2A}{B^2}\int \xi\, d\xi, \qquad \ln \Pi = \frac{A}{B^2}\xi^2 + C$$

Hence,

$$\Pi_{\text{st}}(\xi) \propto \exp\left(\frac{A}{B^2}\xi^2\right)$$

If $A < 0$, the stationary distribution is Gaussian,

$$\xi \sim \text{Gaussian}(0, -\frac{B^2}{2A})$$

4 Covariance Dynamics and the Lyapunov Equation

At an asymptotically stable macroscopic stationary state x^*,

$$r_+(x^*) - r_-(x^*) = 0,$$

and A and B^2 are constants evaluated at x^*. The linear Fokker–Planck equation is then

$$\frac{\partial \Pi}{\partial t} = -A\frac{\partial}{\partial \xi}(\xi \Pi) + \frac{1}{2}B^2\frac{\partial^2 \Pi}{\partial \xi^2} \tag{4.1}$$

For fluctuations about the deterministic trajectory, $\langle \xi \rangle = 0$, the fluctuation variance is

$$C(t) = \langle \xi^2 \rangle = \int_{-\infty}^{\infty} \xi^2 \Pi(\xi, t)\, d\xi$$

Differentiating with respect to time gives

$$\frac{dC}{dt} = \int_{-\infty}^{\infty} \xi^2 \frac{\partial \Pi}{\partial t}\, d\xi$$

Substituting Eq. (4.1),

$$\frac{dC}{dt} = -A\int_{-\infty}^{\infty} \xi^2 \frac{\partial}{\partial \xi}(\xi \Pi)\, d\xi + \frac{B^2}{2}\int_{-\infty}^{\infty} \xi^2 \frac{\partial^2 \Pi}{\partial \xi^2}\, d\xi$$

We assume that Π and its derivatives decay sufficiently rapidly as $|\xi| \to \infty$ that all boundary terms vanish.

For the drift term, integration by parts gives

$$\begin{aligned}\int_{-\infty}^{\infty} \xi^2 \frac{\partial}{\partial \xi}(\xi \Pi)\, d\xi &= \Big[\xi^2(\xi\Pi)\Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} 2\xi(\xi\Pi)\, d\xi \\ &= -2\langle \xi^2 \rangle \\ &= -2C\end{aligned}$$

Therefore the drift contribution is $2AC$.

For the diffusion term, integrate by parts twice:

$$\begin{aligned}\int_{-\infty}^{\infty} \xi^2 \frac{\partial^2 \Pi}{\partial \xi^2}\, d\xi &= \left[\xi^2 \frac{\partial \Pi}{\partial \xi}\right]_{-\infty}^{\infty} - 2\int_{-\infty}^{\infty} \xi \frac{\partial \Pi}{\partial \xi}\, d\xi \\ &= -2\left([\xi\Pi]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \Pi\, d\xi\right) \\ &= 2\end{aligned}$$

since $\int_{-\infty}^{\infty} \Pi\, d\xi = 1$. The diffusion contribution is therefore B^2.

Combining the two terms gives the **covariance dynamics**

$$\frac{dC}{dt} = 2AC + B^2 \tag{4.2}$$

The molecule-number fluctuation is

$$n - \Omega x = \sqrt{\Omega}\,\xi,$$

so the molecule-number variance is

$$\sigma = \Big\langle (n - \langle n \rangle)^2 \Big\rangle = \Omega C.$$

Multiplying Eq. (4.2) by Ω gives

$$\frac{d\sigma}{dt} = 2A\sigma + \Omega B^2$$

At stationarity,

$$2A\sigma + \Omega B^2 = 0$$

This is the one-dimensional form of the **Lyapunov equation**.

5 Generalization to N-Component Reaction Networks

We now extend the previous single species derivation to a general N-component biochemical system. Consider a biochemical system with N chemical components, and R elementary reactions. The molecule numbers and macroscopic concentrations are summarized in the vector $\boldsymbol{n} = (n_1, \ldots, n_N)^T$ and $\boldsymbol{x} = (x_1, \ldots, x_N)^T$, respectively, where $i = 1, \ldots, N$ indexes components.

Let S denotes the $N \times R$ stoichiometric matrix of the reaction network, with integer elements S_{ij}, $i = 1, \ldots, N$ and $j = 1, \ldots, R$. When reaction j occurs, n_i changes to $n_i + S_{ij}$. The columns of the stoichiometric matrix S are obtained directly from the net changes produced by individual reactions. For reaction j, the entry S_{ij} is the change in the number of molecules of species X_i when that reaction occurs once. Equivalently,

$$\begin{aligned} S_{ij} = &\text{ number of molecules of species } X_i \text{ produced in reaction } j \\ &- \text{number of molecules of species } X_i \text{ consumed in reaction } j \end{aligned}$$

The Master equation for the molecule-number probability $p_{\boldsymbol{n}}(t)$ is

$$\frac{\partial p_{\boldsymbol{n}}(t)}{\partial t} = \sum_{j=1}^{R} \left(\prod_{i=1}^{N} \mathcal{E}_i^{-S_{ij}} - 1 \right) \left(R_j p_{\boldsymbol{n}}(t) \right)$$

where $\mathcal{E}_i^{-S_{ij}}$ is the step operator. When it acts on a function $g(X_1, ..., X_i, ..., X_N)$, it removes S_{ij} molecules from X_i, that is, $\mathcal{E}_i^{-S_{ij}} g(X_1, ..., X_i, ..., X_N) = g(X_1, ..., X_i - S_{ij}, ..., X_N)$. Here the operator acts on the whole product $R_j p_{\boldsymbol{n}}$.

Following similar Linear Noise Approximation, we obtain the macroscopic dynamics

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \sum_{j=1}^{R} S_{ij} r_j(\boldsymbol{x}) \qquad \Leftrightarrow \qquad \dot{\boldsymbol{x}} = S\boldsymbol{r}(\boldsymbol{x})$$

and the complete linear Fokker–Planck equation

$$\frac{\partial \Pi}{\partial t} = -\sum_{i,k} A_{ik} \frac{\partial (\xi_k \Pi)}{\partial \xi_i} + \frac{1}{2} \sum_{i,k} [BB^T]_{ik} \frac{\partial^2 \Pi}{\partial \xi_i \partial \xi_k}$$

Here A is the Jacobian of the macroscopic rate equations, and BB^T is the noise-generating diffusion matrix. Both are $N \times N$ matrices:

$$A_{ik} = \sum_{j=1}^{R} S_{ij} \frac{\partial r_j}{\partial x_k} = \frac{\partial (S\boldsymbol{r})_i}{\partial x_k},$$

$$[BB^T]_{ik} = \sum_{j=1}^{R} S_{ij} S_{kj} r_j, \qquad BB^T = S \,\mathrm{diag}(\boldsymbol{r}(\boldsymbol{x})) S^T$$

For nonsingular σ, the stationary solution of the Fokker–Planck equation is a multivariate Gaussian distribution:

$$\Pi_{\mathrm{st}}(\boldsymbol{\xi}) = \frac{\Omega^{N/2}}{(2\pi)^{N/2}\sqrt{\det\sigma}} \exp\left[-\frac{\Omega}{2}\boldsymbol{\xi}^T\sigma^{-1}\boldsymbol{\xi}\right]$$

with zero mean vector $\langle\boldsymbol{\xi}\rangle = 0$ and covariance matrix $\langle\boldsymbol{\xi}\boldsymbol{\xi}^T\rangle = \sigma/\Omega$.

At stationarity, the covariance of Gaussian fluctuations satisfies

$$A\sigma + \sigma A^T + \Omega BB^T = 0 \tag{5.1}$$

Equation (5.1) is the **Lyapunov matrix equation**. It expresses a fluctuation–dissipation balance: the covariance changes generated by the linearized deterministic dynamics, $A\sigma+\sigma A^T$, balance the covariance injected by stochastic reaction events, ΩBB^T .

6 Derivation of Paulsson's Noise Formula

We now derive Paulsson's noise formula (Eq. (1)) starting from a two-component birth–death Markov process.

6.1 Two-Component Birth–Death Process and Master Equation

For a two-component birth–death system with molecular species X_1 and X_2 (for instance messenger RNAs and proteins), whose copy numbers are denoted by n_1 and n_2, respectively. Each component undergoes stochastic production and degradation,

$$\begin{aligned} \varnothing &\xrightarrow{R_1^+(n_1)} X_1, & X_1 &\xrightarrow{R_1^-(n_1)} \varnothing \\ X_1 &\xrightarrow{R_2^+(n_1,n_2)} X_1 + X_2, & X_2 &\xrightarrow{R_2^-(n_1,n_2)} \varnothing \end{aligned}$$

The transition rates $R_i^{\pm}$ specify the rates of individual birth and death events that change the molecule number n_i. Here $R_1^{\pm}(n_1)$ depend only on n_1, whereas $R_2^{\pm}(n_1, n_2)$ depend on both n_1 and n_2. This is an example of dynamic disorder where species X_1 provides the randomly fluctuating environment for X_2, as mRNA fluctuations randomize protein synthesis. The four elementary transitions of the resulting Markov process are:

$$\begin{aligned} \{n_1-1, n_2\} &\xleftarrow{R_1^-(n_1)} \{n_1, n_2\} \xrightarrow{R_1^+(n_1)} \{n_1+1, n_2\}, \\ \{n_1, n_2-1\} &\xleftarrow{R_2^-(n_1,n_2)} \{n_1, n_2\} \xrightarrow{R_2^+(n_1,n_2)} \{n_1, n_2+1\} \end{aligned} \tag{S1}$$

The corresponding chemical Master equation is therefore

$$\begin{aligned} \frac{\partial}{\partial t}p_{n_1n_2} = {} & R_1^+(n_1-1)p_{n_1-1,n_2} + R_1^-(n_1+1)p_{n_1+1,n_2} \\ & + R_2^+(n_1,n_2-1)p_{n_1,n_2-1} + R_2^-(n_1,n_2+1)p_{n_1,n_2+1} \\ & - \Big[R_1^+(n_1) + R_1^-(n_1) + R_2^+(n_1,n_2) + R_2^-(n_1,n_2)\Big]p_{n_1n_2} \end{aligned} \tag{S2}$$

$p_{n_1 n_2}$ is the joint probability of n_1 and n_2. The first four terms are probability inflows to state $\{n_1, n_2\}$; the last term is the outflow from state $\{n_1, n_2\}$. This Master equation is exact for the specified Markov process. Nonlinear rates generally require numerical or expansion methods.

The leading-order terms in the Ω-expansion reproduce the **macroscopic rate equations**

$$\begin{aligned} \frac{\partial x_1}{\partial t} &= r_1^+(x_1) - r_1^-(x_1), \\ \frac{\partial x_2}{\partial t} &= r_2^+(x_1, x_2) - r_2^-(x_1, x_2) \end{aligned} \tag{S3}$$

The next order terms gives the Lyapunov matrix equation, where stationary covariance dynamics follow

$$A\sigma + \sigma A^T + \Omega B B^T = 0 \tag{S4}$$

6.2 Jacobian and Diffusion Matrix

For the two-component birth–death system, set $N = 2$, $R = 4$, and order the rates as birth and death of X_1, followed by birth and death of X_2,

$$\boldsymbol{r} = (r_1^+, r_1^-, r_2^+, r_2^-)^T, \qquad S = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

Restoring $r_1^\pm = r_1^\pm(x_1)$ and $r_2^\pm = r_2^\pm(x_1, x_2)$ gives

$$A = \begin{pmatrix} \dfrac{\partial(r_1^+ - r_1^-)}{\partial x_1} & 0 \\ \dfrac{\partial(r_2^+ - r_2^-)}{\partial x_1} & \dfrac{\partial(r_2^+ - r_2^-)}{\partial x_2} \end{pmatrix}, \qquad BB^T = \begin{pmatrix} r_1^+ + r_1^- & 0 \\ 0 & r_2^+ + r_2^- \end{pmatrix}$$

For this particular process, stationarity implies $r_i^+ = r_i^- = r_i$, $i = 1, 2$. The following two sections express these coefficients in terms of averages of numbers of molecules, lifetimes and logarithmic gains.

6.3 Lifetimes and Logarithmic Gains

At the fixed point, define

$$H_{ij} = \left. \frac{\partial \ln(r_i^-/r_i^+)}{\partial \ln x_j} \right|_{x^*}$$

Parameter H_{ij} summarizes how a change in x_j affects the balance between synthesis and degradation of x_i. The Jacobian A can be written as

$$A_{ij} = \frac{\partial}{\partial x_j}\left(\frac{\partial x_i}{\partial t}\right) = \frac{r_i^+}{x_j}\frac{\partial \ln r_i^+}{\partial \ln x_j} - \frac{r_i^-}{x_j}\frac{\partial \ln r_i^-}{\partial \ln x_j} = -\frac{r_i}{x_j}\frac{\partial \ln(r_i^-/r_i^+)}{\partial \ln x_j} \equiv -\frac{r_i}{x_j}H_{ij} \tag{S5}$$

The average molecular lifetime τ_i can be defined by the concentration divided by the total rate of formation or elimination $\tau_i = x_i/r_i$. Here as A and $\Omega B B^T$ are evaluated at an asymptotically stable macroscopic steady state where all x_i/r_i are constant, each molecule is

then surrounded by a constant environment of other molecules, and the lifetime definitions are both exact and universal, resulting in

$$\Omega r_i = \frac{\langle n_i \rangle}{\tau_i}, \tag{S6}$$

Equation (S5) can then be rewritten as

$$A_{ij} = -\frac{x_i}{x_j}\frac{r_i}{x_i}H_{ij} = -\frac{\langle n_i \rangle}{\langle n_j \rangle}\frac{H_{ij}}{\tau_i} \tag{S7}$$

For the particular process considered here, the diffusion matrix BB^T similarly follows

$$(\Omega BB^T)_{ii} = 2\Omega r_i = \frac{2\langle n_i \rangle}{\tau_i}, \qquad (\Omega BB^T)_{ij} = 0 \quad (i \neq j) \tag{S8}$$

For an activating input that increases r_2^+, $H_{21} < 0$. For a repressive input, $H_{21} > 0$. The sign convention follows the ratio 'elimination/production' in the definition of H_{ij}.

6.4 Solving the Normalized Covariance Equation

We introduce the normalized matrices:

$$V_{ij} = \frac{\sigma_{ij}}{\langle n_i \rangle \langle n_j \rangle} = V_{ji}, \qquad M_{ij} = A_{ij}\frac{\langle n_j \rangle}{\langle n_i \rangle}, \qquad D_{ij} = \frac{\Omega(BB^T)_{ij}}{\langle n_i \rangle \langle n_j \rangle} = D_{ji} \tag{S9}$$

Then $V_{ii} = \sigma_i^2/\langle n_i \rangle^2$ and Eq. (S4) can be written as

$$MV + (MV)^T + D = 0 \tag{S10}$$

Equations (S7)–(S8) give

$$M = \begin{pmatrix} -H_{11}/\tau_1 & 0 \\ -H_{21}/\tau_2 & -H_{22}/\tau_2 \end{pmatrix}, \quad D = \begin{pmatrix} 2/(\langle n_1 \rangle \tau_1) & 0 \\ 0 & 2/(\langle n_2 \rangle \tau_2) \end{pmatrix} \tag{S11}$$

Solving Eq. (S10) for V gives

$$V_{11} = \frac{1}{\langle n_1 \rangle H_{11}} \tag{S12}$$

$$V_{12} = -V_{11}\frac{H_{21}}{H_{22}}\frac{H_{22}/\tau_2}{H_{11}/\tau_1 + H_{22}/\tau_2} = V_{21} \tag{S13}$$

$$\begin{aligned} V_{22} &= \frac{1}{\langle n_2 \rangle H_{22}} - \frac{H_{21}}{H_{22}}V_{12} \\ &= \frac{1}{\langle n_2 \rangle H_{22}} + V_{11}\frac{H_{21}^2}{H_{22}^2}\frac{H_{22}/\tau_2}{H_{11}/\tau_1 + H_{22}/\tau_2} \end{aligned} \tag{S14}$$

Restoring the original parameters yields Paulsson's equation (1), the stationary fluctuations around a stable fixed point:

$$\frac{\sigma_2^2}{\langle n_2 \rangle^2} \simeq \underbrace{\frac{1}{\langle n_2 \rangle H_{22}}}_{\text{intrinsic noise}} + \underbrace{\frac{\sigma_1^2}{\langle n_1 \rangle^2}}_{n_1 \text{ noise}} \underbrace{\left(\frac{H_{21}}{H_{22}}\right)^2}_{\text{Susceptibility}} \underbrace{\frac{H_{22}/\tau_2}{H_{11}/\tau_1 + H_{22}/\tau_2}}_{\text{Time-averaging}} \tag{1}$$

The first term is intrinsic noise of X_2; the product of the last three factors is the extrinsic contribution from X_1. The approximate equality refers to the LNA.

6.5 Interpretation of Intrinsic and Extrinsic Noise

Intrinsic noise depends on the average number of molecules and how systematic adjustments (rate H_{22}/τ_2) quench spontaneous fluctuations (rate $1/\tau_2$). The normalized adjustment rate H_{22} can also be interpreted as the statistical bias to return to the average rather than deviate further.

Extrinsic noise instead depends on the magnitude of n_1 fluctuations and how strongly n_1 affects n_2.

Susceptibility: The normalized susceptibility factor H_{21}/H_{22} reflects that a 1% increase in n_1 gives a H_{21} per cent increase in R_2^-/R_2^+; which makes n_2 adjust towards a H_{21}/H_{22} per cent lower average quasi-steady-state.

Time-averaging: When n_1 changes rapidly (high H_{11}/τ_1) or n_2 adjusts slowly (low H_{22}/τ_2), n_2 does not have time to reach its quasi-steady-state before n_1 changes anew. Consecutive ups and downs in n_1 then cancel out and n_2 time-averages over the recent history of n_1 fluctuations.

With this distinction between intrinsic and extrinsic contributions in mind, we can now see why Paulsson formulates equation (1) in terms of relative variance. For intrinsic noise it is convenient to use $\sigma_2^2/\langle n_2\rangle$ for a size-independent comparison, but this artificially forces extrinsic noise to increase with $\langle n_2\rangle$, as in $\sigma_2^2/\langle n_2\rangle \approx H_{22}^{-1} + E\langle n_2\rangle$, where E is the second term in equation (1). Unless the measure matches the noise, scale artefacts may thus completely distort interpretations of dynamics. For instance, if all protein (X_2) noise came from fluctuations in protease levels, $\sigma_2^2/\langle n_2\rangle^2$ would typically be independent of transcription and translation rates. But because both of these processes affect $\langle n_2\rangle$, measuring noise strength by $\sigma_2^2/\langle n_2\rangle$ would make it appear as if they contributed noise, even though they have nothing to do with either its production or transmission.

6.6 Autoregulation and Noise Propagation

The basic principles of noise in genetic networks can be understood by applying equation (1) to the central dogma of molecular biology. To suppress the noise in systems such as DNA replication, transcription, or translation, cells commonly use autorepression that increases and decreases synthesis at low and high concentrations respectively.

Autorepression can raise the effective H_{22}, and thus:

(1) increase the adjustment rate H_{22}/τ_2 relative to the rate $1/\tau_2$ of spontaneous randomization and thereby suppress intrinsic noise around a given average number of molecules;

(2) increase the adjustment rate H_{22}/τ_2 relative to the rate H_{11}/τ_1 of environmental changes and thereby amplify extrinsic noise by preventing time averaging;

(3) decrease the susceptibility H_{21}/H_{22} of the quasi steady-state, which typically overcompensates for the impaired time-averaging and produces a net decrease in extrinsic noise.

This may explain the popularity of autorepression in transcription networks and its ubiquity in replication control of chromosomes and plasmids where it has been similarly described. Equation (1) thus unifies models of autoreplication, constitutive transcription and translation, and the stabilizing effect of autorepression, all in disordered environments.

References

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